Worksheet 1 Physics Grade 10

Exercise 1: Relative Motion

Two passengers A and B are sitting in front of each other in a moving bus. Let C be a man who is stationary on the road.

Tell, indicating the reference in each case, whether :

1) A is in motion or at rest with respect to B.
2) A is in motion or at rest with respect to C.
3) C is in motion or at rest with respect to the Earth.
4) C is in motion or at rest with respect to B.

Exercise 2: Recording a Motion

The following record shows, at real scale, the positions occupied by a mobile (A). The motion of (A) is related to an x'x axis, of origin O, coincided with A_0, and of unit vector $\vec{i}$ oriented in the direction of motion.

Given:
$A_0A_1 = 0.4$ cm ; $A_1A_2 = 1.2$ cm ; $A_2A_3 = 2$ cm ; $A_3A_4 = 2.8$ cm ; $A_4A_5 = 3.6$ cm ; $A_5A_6 = 4.4$ cm.

1) What is the nature of the trajectory of (A)? Justify.
2) The origin of time is taken when the mobile passes through O ($A_0 \equiv O$). What is the abscissa of the mobile at t_0, t_1, t_2, t_3, t_4, t_5 et t_6.
3) Determine the displacement of the mobile during the duration
   3-1) $\Delta t = (t_1 - t_0)$.
   3-2) $\Delta t = (t_5 - t_1)$
   3-3) $\Delta t = (t_6 - t_2)$